Course Code: SKL2711	**Course Name:** Corporate Training and Career Skills	
Credits: 2 (0-2-0)	**Contact Hours:** 2 hours Discussion(Theory/Practical)	
Batch: 2024, 4th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
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Aim of the course: The main objective of this course is to introduce students to different data structures and illustrate their effective use in solving technical and logical problems. The course comprehensively explores different problem-solving techniques and skills. Proficiency in problem-solving skills is a fundamental expectation for any competent developer, as these concepts are commonly assessed by reputable companies during the screening process for software developer positions.

Course Overview and Context: The primary emphasis will be on achieving a deep understanding of data structures, their implementation, practical applications through problem-solving scenarios, exploring various programming paradigms, algorithm analysis, and the practical application of different data structures and algorithms. This course explores the fundamental workings of algorithms and data structures, which lie at its essence.

Course Outcomes (CO): At the end of the course, the students should be able to do the following:

CO	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1	To understand the core principles underlying data structures and their significance in problem-solving.	**Employability, Knowledge Enhancement**	→ Intensive theory lectures on fundamental data structures (arrays, stacks, queues, linked lists, trees, graphs). → In-class problem-solving discussions and concept quizzes. → Visualisation-based demonstrations of data structure operations. → Case-based explanations linking data structures to real-world software systems.
CO2	To apply various data structures effectively in solving specific technical and logical problems.	**Skill Development, Employability**	→ Hands-on laboratory sessions implementing data structures using a programming language (C/C++/Java/Python). → Coding assignments focused on algorithmic problem-solving. → Mini programming tasks involving selection and implementation of appropriate data structures. → Continuous assessment through lab evaluations and practical tests.

CO3	To analyze and compare the performance of different data structures in specific problem-solving scenarios.	**Entrepreneurship, Skill Development**	→ Comparative analysis of time and space complexity of data structures. → Problem-based assignments requiring justification of data structure choice.

Prerequisite: if any

1. Basic Structure of Data Structures and Fundamentals of Programming

CO PO Mapping : Three levels of mapping (1: weak, 2: medium, 3: strong)

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1		1			1								1		3	
CO2	3	2	3		3								3		2	
CO3		3			2								3		2	

Learning Resources:

MOOCs

1. https://www.coursera.org/specializations/data-structures-algorithms
2. https://nptel.ac.in/courses/106102064
3. https://leetcode.com/explore/interview/card/leetcodes-interview-crash-course-data-structures-and-algorithms/

Experiential Learning Component (80 %):

In this course, students attempt to develop a practical application to address real world problem by proposing novel improvements to an existing method, applying existing methods to different types of problems. The students are also expected to implement and show the results of the proposed solution or attempt to reimplement and improve on a research paper on a topic of their choice. Students can also integrate methods/techniques of other courses such as Digital Image Processing, Artificial Intelligence, etc. A separate examination will be conducted to evaluate the solution provided by each student in such cases. This attempts to provide the following skills:

- Thinking critically to deal with uncertainty and ambiguity.
- Application of for handling real-world problems.
- skills in writing technical concepts in the form of research papers.

Assessment Pattern: The final grade will be determined by the marks earned during different evaluations conducted during the semester. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule / Timeline	Rubrics / Remarks	CO Mapping
Class Participation (Experiential Learning)	**10%**	**Continuous**	- Based on regular participation in in-class problem solving and discussions. - Timely submission of assigned MAANG-level coding problems. - Active engagement in solution optimization and peer discussions.	CO1, CO2
Evaluation – 1 (Experiential Learning)	**25%**	**Week of Mid Term Examination**	- 90-minute coding evaluation on an online platform. - Focus on implementation of data structures, algorithmic correctness, and optimization. - Emphasis on handling edge cases and time–space complexity.	CO2, CO3
Evaluation – 2 (Experiential Learning)	**25%**	**Last week of April 2026**	- 90-minute advanced coding evaluation on an online platform. - MAANG-level problem statements involving trees, graphs, DP, and maps. - Evaluation based on efficiency, scalability, and quality of solution.	CO2, CO3
End Term Evaluation	**40%**	**Last week of semester / As per academic calendar**	- Comprehensive coding-based assessment conducted on an online platform. - Problems focused on real interview scenarios, optimisation techniques, and complex data structure applications - No theory-only questions; evaluation strictly based on coding and logic.	CO1, CO2, CO3

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.
Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.
Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.
Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.
Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.